

Quantum Vertical Motion using Airy Functions

A PROJECT REPORT

Soumy Mittal (DL.AI.U4AID25095)

in partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY (B.Tech) IN ARTIFICIAL INTELLIGENCE AND DATA SCIENCE (AIDS)

Under the guidance of

Dr. Mrityunjay Guha Majumdar

and

Dr. Rakesh Kumar

Submitted to



AMRITA VISHWA VIDYAPEETHAM

AMRITA SCHOOL OF ARTIFICIAL INTELLIGENCE

FARIDABAD – 121002, June 2026

Abstract

This report presents a rigorous mathematical study of vertical projectile dynamics under gravity, analysed through the lens of quantum mechanics, computational linear algebra, and probabilistic intelligent-system modelling. The governing physical equation is the time-independent Schrödinger equation with a linear gravitational potential, whose analytical solutions are Airy functions. The continuous wavefunction is discretised into a finite-dimensional state vector, and Fourier Transformation is implemented to go from position space to momentum space.

The squared wavefunction is reinterpreted as a probability density function, transforming the particle's vertical position into a continuous random variable. Statistical measures—expectation value, variance, and standard deviation—are computed to characterise the uncertainty in the particle's position across different energy eigenstates. Finally, an optimisation framework is developed to identify the energy configuration that maximises the probability density at a prescribed target height, unifying quantum mechanics, linear algebra, and stochastic analysis in a single coherent model.

The project demonstrates a strong interdisciplinary connection between quantum physics, eigenvalue analysis, Fourier Transform, and probabilistic modelling paradigms central to modern intelligent systems.

Keywords: Schrödinger equation, Airy functions, gravitational quantum states, wavefunction discretisation, Fourier Transformation, random variable, expectation value, probabilistic optimisation, computational mechanics, intelligent systems.

Contents

Abstract	1
1 Introduction	7
1.1 Motivation and Context	7
1.2 Project Overview and Objectives	7
2 The Schrödinger Equation under Gravitational Potential	9
2.1 Physical Setup	9
2.2 Governing Differential Equation	9
2.3 Transformation of the Schrödinger Equation into the Airy Equation	10
3 Airy Functions and Quantised Energy Eigenstates	14
3.1 General Solution of the Airy Equation	14
3.2 Final Wavefunction in Terms of y	15
3.3 Quantisation Condition and Energy Eigenvalues	15
3.4 Wavefunction Normalisation	17
3.5 Classically Forbidden Region	20
3.5.1 Classically Allowed Region	20
3.5.2 Classically Forbidden Region	21
4 Linear Algebra: Discretisation and Fourier Transformation	23
4.1 Discretization of the Wavefunction and Computational Eigenvalue Analysis	23
4.1.1 Spatial Discretization	23
4.1.2 Finite Difference Approximation	24
4.1.3 Matrix Formulation	24
4.1.4 Computational Eigenvalue Problem	25
4.1.5 Comparison with Analytical Energy Eigenvalues	26
4.1.6 Physical Interpretation	26

4.2	Momentum-Space Analysis	27
5	Probability Theory and Random Variables	29
5.1	Random Variable Interpretation of the Quantum Gravitational System . .	29
5.1.1	Probability Density Function	30
5.1.2	Physical Interpretation of the Probability Density	30
5.1.3	Expectation Value of Position	31
5.1.4	Variance of Position	31
5.1.5	Relation to the Probability Density Graphs	32
5.1.6	Connection to Quantum Mechanics and Computational Analysis . .	32
6	Optimisation: Uncertainty Minimization	34
6.1	Optimization Through Uncertainty Minimization	34
6.1.1	Optimization Objective	35
6.1.2	Determination of the Most Probable Height	35
6.1.3	Physical Interpretation	36
6.1.4	Optimization Results	36
7	Results and Discussion	38
7.1	Introduction	38
7.2	Airy Wavefunctions	38
7.3	Probability Density Distributions	39
7.4	Comparison of Analytical and Computational Energy Eigenvalues	40
7.5	Expectation Values of Position	41
7.6	Variance and Positional Uncertainty	42
7.7	Classically Allowed and Forbidden Regions	42
7.8	Computational Eigenvalue Analysis	43
7.9	Connection to Computational Mechanics and Intelligent Systems	44
7.10	Summary of Results	44
8	Conclusions	46
8.1	Summary of Mathematical Themes	46
8.2	Key Conclusions	46
8.3	Final Characterisation	47
A	MATLAB Code Listings	48
A.1	Wavefunction Computation, Fourier Transform Random Variables and Optimization	48
A.2	Wavefunction Matrix	54

List of Symbols

57

List of Figures

3.1	Normalized Airy-function wavefunctions for the first three quantum states of the gravitational system. The wavefunctions exhibit oscillatory behavior in the classically allowed region and exponential decay in the classically forbidden region. Higher-energy states contain larger numbers of oscillations and extend to greater heights against gravity.	20
4.1	Fourier transform of the Airy-function wavefunction.	28
6.1	Uncertainty minimization using variance. The probability density functions of the first three quantum states are shown. The red dashed line indicates the most probable height corresponding to the minimum-variance state. The red marker identifies the optimal particle location. State 1 was found to possess the smallest variance and therefore represents the most spatially localized quantum state.	37
7.1	Probability density functions for the first three quantum states of the gravitational system.	39
7.2	Comparison between analytical Airy-function energy eigenvalues and computational eigenvalues obtained from the discretized Hamiltonian matrix. The close agreement between the two sets of results validates the numerical finite difference implementation.	41

List of Tables

2.1	Physical parameters used throughout the report.	13
3.1	First five zeros of $\text{Ai}(z)$ and the corresponding dimensionless energy eigenvalues.	16
6.1	Results of Uncertainty Minimization	36
7.1	Comparison of Numerical and Analytical Energy Eigenvalues	40
8.1	Mathematical themes and their roles in the project.	46

Introduction

1.1 Motivation and Context

Classical mechanics describes the trajectory of a projectile under gravity as a perfectly deterministic path. In the quantum regime, however, the concept of a *definite* trajectory breaks down: a particle's position is described by a probability amplitude governed by the Schrödinger equation. This tension between classical determinism and quantum uncertainty provides both the physical motivation and the mathematical richness of the present project.

The intersection of quantum mechanics with computational tools—eigenvalue decomposition, fourier transformation, and probabilistic modelling—lies at the heart of modern *Mathematics of Intelligent Systems*. Neural networks, signal-processing pipelines, and probabilistic inference algorithms all rely on the same linear-algebraic and statistical machinery employed here in a quantum-mechanical context.

1.2 Project Overview and Objectives

The project is structured around five interconnected mathematical themes:

1. **Differential Equations** — formulation and solution of the Schrödinger equation under a linear gravitational potential.
2. **Quantum Mechanics** — derivation of quantised energy eigenstates and Airy-function wavefunctions.
3. **Linear Algebra & Fourier Transform** — discretisation of the wavefunction into a matrix structure and Fourier transform for space transformation.

4. **Probability & Random Variables** — reinterpretation of the wavefunction modulus squared as a probability density function, computation of statistical moments, and stochastic analysis of positional uncertainty.
5. **Optimisation** — Finding the optimal/best energy state by minimizing the variance.

The Schrödinger Equation under Gravitational Potential

2.1 Physical Setup

Consider a particle of mass m confined to the half-space $y \geq 0$ in the presence of a uniform gravitational field. The potential energy is:

$$V(y) = \begin{cases} mgy, & y \geq 0, \\ \infty, & y < 0, \end{cases} \quad (2.1)$$

where $g = 9.8 \text{ m s}^{-2}$ is the standard gravitational acceleration. The impenetrable floor at $y = 0$ enforces the boundary condition $\psi(0) = 0$. The wavefunction must also vanish at infinity:

$$\lim_{y \rightarrow \infty} \psi(y) = 0. \quad (2.2)$$

2.2 Governing Differential Equation

The time-independent Schrödinger equation for the system is:

Schrödinger Equation — Gravitational Potential

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi}{dy^2} + mgy \psi(y) = E \psi(y), \quad y \in [0, \infty), \quad (2.3)$$

where $\hbar = h/(2\pi)$ is the reduced Planck constant and E is the (unknown) energy eigenvalue.

Rearranging eq. (2.3):

$$\frac{d^2\psi}{dy^2} = \frac{2m}{\hbar^2}(mgy - E)\psi(y). \quad (2.4)$$

2.3 Transformation of the Schrödinger Equation into the Airy Equation

Starting from the time-independent Schrödinger equation under the gravitational potential:

$$V(y) = mgy$$

the governing equation becomes:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dy^2} + mgy\psi(y) = E\psi(y)$$

Rearranging the equation:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dy^2} = (E - mgy)\psi(y)$$

Multiplying both sides by:

$$-\frac{2m}{\hbar^2}$$

gives:

$$\frac{d^2\psi}{dy^2} = \frac{2m}{\hbar^2}(mgy - E)\psi(y)$$

Expanding the expression:

$$\frac{d^2\psi}{dy^2} = \left(\frac{2m^2g}{\hbar^2}y - \frac{2mE}{\hbar^2} \right) \psi(y)$$

Factoring out the common term:

$$\frac{d^2\psi}{dy^2} = \frac{2m^2g}{\hbar^2} \left(y - \frac{E}{mg} \right) \psi(y)$$

Define the classical turning point as:

$$y_0 = \frac{E}{mg}$$

Substituting into the equation:

$$\frac{d^2\psi}{dy^2} = \frac{2m^2g}{\hbar^2}(y - y_0)\psi(y)$$

To convert the equation into standard Airy form, introduce the dimensionless variable:

$$z = \left(\frac{2m^2g}{\hbar^2}\right)^{1/3} (y - y_0)$$

Let:

$$\alpha = \left(\frac{2m^2g}{\hbar^2}\right)^{1/3}$$

Then:

$$z = \alpha(y - y_0)$$

Using the chain rule:

$$\frac{d}{dy} = \frac{dz}{dy} \frac{d}{dz}$$

Since:

$$\frac{dz}{dy} = \alpha$$

we obtain:

$$\frac{d}{dy} = \alpha \frac{d}{dz}$$

Differentiating again:

$$\frac{d^2}{dy^2} = \alpha^2 \frac{d^2}{dz^2}$$

Substituting into the transformed Schrödinger equation:

$$\alpha^2 \frac{d^2 \psi}{dz^2} = \frac{2m^2 g}{\hbar^2} (y - y_0) \psi$$

From:

$$z = \alpha(y - y_0)$$

we get:

$$y - y_0 = \frac{z}{\alpha}$$

Substituting:

$$\alpha^2 \frac{d^2 \psi}{dz^2} = \frac{2m^2 g}{\hbar^2} \frac{z}{\alpha} \psi$$

Since:

$$\alpha^3 = \frac{2m^2 g}{\hbar^2}$$

the equation becomes:

$$\alpha^2 \frac{d^2 \psi}{dz^2} = \frac{\alpha^3}{\alpha} z \psi$$

which simplifies to:

$$\alpha^2 \frac{d^2 \psi}{dz^2} = \alpha^2 z \psi$$

Cancelling α^2 from both sides:

$$\frac{d^2 \psi}{dz^2} = z \psi$$

Rearranging:

$$\frac{d^2\psi}{dz^2} - z\psi = 0$$

This is the standard Airy differential equation.

Table 2.1: Physical parameters used throughout the report.

Symbol	Quantity	Value	Units
$\hbar$	Reduced Planck constant	1.055×10^{-34}	J·s
m	Particle mass	1.0×10^{-3}	kg
g	Gravitational acceleration	9.8	m s ⁻²

Airy Functions and Quantised Energy Eigenstates

3.1 General Solution of the Airy Equation

The Airy differential equation is given by:

$$\frac{d^2\psi}{dz^2} - z\psi = 0$$

The equation has two independent solutions:

$$\psi(z) = C_1 Ai(z) + C_2 Bi(z)$$

where:

- $Ai(z)$ = Airy function of the first kind
- $Bi(z)$ = Airy function of the second kind

For large positive values of z , the Airy function $Bi(z)$ diverges exponentially. Since a physically acceptable quantum wavefunction must remain finite and normalizable, the $Bi(z)$ term is discarded.

Thus, the physically acceptable solution becomes:

$$\psi(z) = C Ai(z)$$

3.2 Final Wavefunction in Terms of y

Substituting the transformed variable:

$$z = \left(\frac{2m^2g}{\hbar^2} \right)^{1/3} \left(y - \frac{E}{mg} \right)$$

the final wavefunction is obtained as:

$$\psi(y) = C \operatorname{Ai} \left[\left(\frac{2m^2g}{\hbar^2} \right)^{1/3} \left(y - \frac{E}{mg} \right) \right]$$

This represents the Airy-function solution of the Schrödinger equation for a quantum particle moving vertically under the influence of gravity.

3.3 Quantisation Condition and Energy Eigenvalues

The physically acceptable Airy-function solution is:

$$\psi(y) = C \operatorname{Ai} \left[\left(\frac{2m^2g}{\hbar^2} \right)^{1/3} \left(y - \frac{E}{mg} \right) \right]$$

For the quantum bouncing-ball system, an infinite potential wall is present at:

$$y = 0$$

Therefore, the boundary condition is:

$$\psi(0) = 0$$

Substituting $y = 0$ into the wavefunction:

$$\psi(0) = C \operatorname{Ai} \left[\left(\frac{2m^2g}{\hbar^2} \right)^{1/3} \left(0 - \frac{E}{mg} \right) \right]$$

Simplifying:

$$\psi(0) = C \operatorname{Ai} \left(-\alpha \frac{E}{mg} \right)$$

where:

$$\alpha = \left(\frac{2m^2g}{\hbar^2} \right)^{1/3}$$

Applying the boundary condition:

$$\psi(0) = 0$$

gives:

$$C \operatorname{Ai} \left(-\alpha \frac{E}{mg} \right) = 0$$

Since $C \neq 0$ for a nontrivial solution, the condition becomes:

$$\operatorname{Ai} \left(-\alpha \frac{E}{mg} \right) = 0$$

The zeros of the Airy function determine the allowed energy states.

Let:

$$a_n$$

represent the positive zeros of the Airy function:

Table 3.1: First five zeros of $\operatorname{Ai}(z)$ and the corresponding dimensionless energy eigenvalues.

State n	Zero a_n	$E_n/(mg\ell)$
1	-2.3381	2.3381
2	-4.0879	4.0879
3	-5.5206	5.5206
4	-6.7867	6.7867
5	-7.9441	7.9441

Thus:

$$Ai(-a_n) = 0$$

Comparing with the quantization condition:

$$Ai\left(-\alpha \frac{E_n}{mg}\right) = 0$$

we obtain:

$$-\alpha \frac{E_n}{mg} = -a_n$$

which simplifies to:

$$\alpha \frac{E_n}{mg} = a_n$$

Therefore:

$$E_n = \frac{mg}{\alpha} a_n$$

Substituting:

$$\alpha = \left(\frac{2m^2g}{\hbar^2}\right)^{1/3}$$

gives the quantized energy eigenvalues:

$$E_n = a_n \left(\frac{\hbar^2 mg^2}{2}\right)^{1/3}$$

Thus, the energy of the quantum particle becomes quantized due to the boundary condition imposed by the infinite potential wall.

3.4 Wavefunction Normalisation

The physically acceptable Airy-function wavefunction is:

$$\psi_n(y) = C_n Ai(\alpha y - a_n)$$

where:

$$\alpha = \left(\frac{2m^2g}{\hbar^2} \right)^{1/3}$$

and a_n are the zeros of the Airy function.

For a physically meaningful quantum state, the wavefunction must satisfy the normalization condition:

$$\int_0^\infty |\psi_n(y)|^2 dy = 1$$

Substituting the wavefunction:

$$\int_0^\infty |C_n Ai(\alpha y - a_n)|^2 dy = 1$$

Since C_n is constant:

$$C_n^2 \int_0^\infty Ai^2(\alpha y - a_n) dy = 1$$

Introduce the variable transformation:

$$x = \alpha y - a_n$$

Differentiating:

$$dx = \alpha dy$$

Thus:

$$dy = \frac{dx}{\alpha}$$

The integration limits become:

$$y = 0 \quad \Rightarrow \quad x = -a_n$$

$$y \rightarrow \infty \quad \Rightarrow \quad x \rightarrow \infty$$

Substituting into the normalization integral:

$$C_n^2 \int_{-a_n}^{\infty} Ai^2(x) \frac{dx}{\alpha} = 1$$

or:

$$\frac{C_n^2}{\alpha} \int_{-a_n}^{\infty} Ai^2(x) dx = 1$$

Using the Airy-function identity:

$$\int_{-a_n}^{\infty} Ai^2(x) dx = [Ai'(-a_n)]^2$$

the equation becomes:

$$\frac{C_n^2}{\alpha} [Ai'(-a_n)]^2 = 1$$

Solving for the normalization constant:

$$C_n^2 = \frac{\alpha}{[Ai'(-a_n)]^2}$$

Taking square root:

$$C_n = \frac{\sqrt{\alpha}}{|Ai'(-a_n)|}$$

Therefore, the normalized wavefunction is:

$$\psi_n(y) = \frac{\sqrt{\alpha}}{|Ai'(-a_n)|} Ai(\alpha y - a_n)$$

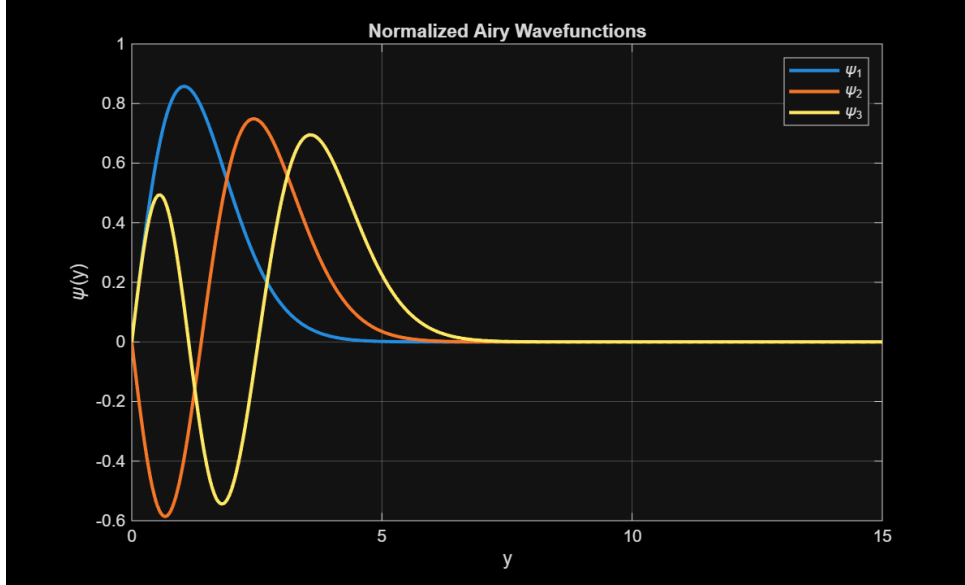


Figure 3.1: Normalized Airy-function wavefunctions for the first three quantum states of the gravitational system. The wavefunctions exhibit oscillatory behavior in the classically allowed region and exponential decay in the classically forbidden region. Higher-energy states contain larger numbers of oscillations and extend to greater heights against gravity.

Thus, the wavefunction is normalized such that the total probability of finding the particle in the allowed region is equal to unity.

3.5 Classically Forbidden Region

For a particle moving under the gravitational potential:

$$V(y) = mgy$$

the total energy of the particle is given by E_n .

The motion of the particle can be divided into two regions based on the comparison between the total energy and the potential energy.

3.5.1 Classically Allowed Region

The region where:

$$E_n > V(y)$$

or equivalently,

$$E_n > mgy$$

is known as the classically allowed region.

In this region, the particle possesses sufficient energy to exist and move classically. The Airy wavefunction exhibits oscillatory behavior in this region, representing standing-wave-like quantum states.

The classical turning point occurs at:

$$y_t = \frac{E_n}{mg}$$

where:

$$E_n = mgy_t$$

At this point, the kinetic energy of the particle becomes zero.

3.5.2 Classically Forbidden Region

The region where:

$$V(y) > E_n$$

or:

$$mgy > E_n$$

is known as the classically forbidden region.

According to classical mechanics, the particle cannot enter this region because the potential energy exceeds the total energy of the particle.

However, in quantum mechanics, the wavefunction does not abruptly vanish at the turning point. Instead, the Airy-function solution extends slightly into the forbidden region and decays exponentially.

For large positive values of z , the Airy function behaves as:

$$Ai(z) \sim \frac{1}{2\sqrt{\pi}} z^{-1/4} e^{-2z^{3/2}/3}$$

Thus, the wavefunction gradually approaches zero as the height increases.

Consequently, the probability density:

$$|\psi(y)|^2$$

also decreases exponentially in the forbidden region.

This indicates that the probability of finding the particle at very large heights becomes extremely small.

The exponential decay of the wavefunction in the classically forbidden region is a purely quantum-mechanical phenomenon and demonstrates the probabilistic nature of the particle under the gravitational potential.

Linear Algebra: Discretisation and Fourier Transformation

4.1 Discretization of the Wavefunction and Computational Eigenvalue Analysis

The analytical Airy-function solution provides the exact quantum states of the gravitational system. In order to study the system computationally, the Schrödinger equation was discretized using the Finite Difference Method (FDM).

The time-independent Schrödinger equation under gravity is:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(y)}{dy^2} + mgy\psi(y) = E\psi(y)$$

4.1.1 Spatial Discretization

The spatial domain was divided into equally spaced grid points:

$$y_1, y_2, y_3, \dots, y_N$$

with grid spacing:

$$h = \frac{L}{N + 1}$$

where:

- L = maximum height of the computational domain
- N = number of discretization points

The wavefunction was approximated at discrete points:

$$\psi(y) \rightarrow \psi_i$$

where:

$$\psi_i = \psi(y_i)$$

4.1.2 Finite Difference Approximation

The second derivative in the Schrödinger equation was approximated using the central finite difference formula:

$$\frac{d^2\psi}{dy^2} \approx \frac{\psi_{i+1} - 2\psi_i + \psi_{i-1}}{h^2}$$

Substituting this approximation into the Schrödinger equation gives:

$$-\frac{\hbar^2}{2m} \frac{\psi_{i+1} - 2\psi_i + \psi_{i-1}}{h^2} + mgy_i\psi_i = E\psi_i$$

Rearranging:

$$-\frac{\hbar^2}{2mh^2}\psi_{i+1} + \left(\frac{\hbar^2}{mh^2} + mgy_i\right)\psi_i - \frac{\hbar^2}{2mh^2}\psi_{i-1} = E\psi_i$$

4.1.3 Matrix Formulation

The discretized equations for all grid points were assembled into matrix form:

$$H\Psi = E\Psi$$

where:

- H = Hamiltonian matrix
- Ψ = discretized wavefunction vector

- E = energy eigenvalue

The wavefunction vector is written as:

$$\Psi = \begin{bmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \vdots \\ \psi_N \end{bmatrix}$$

The Hamiltonian matrix becomes a tridiagonal matrix of the form:

$$H = \begin{bmatrix} d_1 & c & 0 & 0 & \cdots \\ c & d_2 & c & 0 & \cdots \\ 0 & c & d_3 & c & \cdots \\ 0 & 0 & c & d_4 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

where:

$$c = -\frac{\hbar^2}{2mh^2}$$

and

$$d_i = \frac{\hbar^2}{mh^2} + mgy_i$$

4.1.4 Computational Eigenvalue Problem

The matrix equation:

$$H\Psi_n = E_n\Psi_n$$

was solved numerically using MATLAB eigenvalue solvers.

The computed eigenvalues:

$$E_1, E_2, E_3, \dots$$

represent the quantized energy levels of the system, while the eigenvectors:

$$\Psi_1, \Psi_2, \Psi_3, \dots$$

represent the discretized quantum wavefunctions.

4.1.5 Comparison with Analytical Energy Eigenvalues

The analytical Airy-function solution produced the exact quantized energies:

$$E_n = a_n \left(\frac{\hbar^2 m g^2}{2} \right)^{1/3}$$

where a_n are the zeros of the Airy function.

The numerical eigenvalues obtained from the discretized Hamiltonian matrix were compared with these analytical energy values.

The comparison demonstrated that the computational eigenvalues converged closely to the analytical Airy-function energies as the grid spacing became smaller and the number of discretization points increased.

This agreement validated the computational finite difference formulation and confirmed that the discretized Hamiltonian matrix successfully reproduced the quantized gravitational states predicted analytically.

4.1.6 Physical Interpretation

The computational formulation transformed the continuous Schrödinger differential equation into a matrix eigenvalue problem.

In this formulation:

- Eigenvalues correspond to the allowed quantum energies
- Eigenvectors correspond to the discretized wavefunctions
- Higher eigenstates exhibit more oscillatory behavior and larger spatial spread

Thus, the discretization process established a direct connection between:

- Quantum mechanics
- Computational mechanics

- Linear algebra
- Eigenvalue analysis

allowing the gravitational quantum system to be studied numerically through matrix methods.

4.2 Momentum-Space Analysis

The normalized Airy wavefunction describes the quantum state in position space. To analyze the momentum distribution of the particle, the wavefunction was transformed into momentum space using the Fourier transform.

The momentum-space wavefunction is defined as

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int_0^\infty \psi(y) e^{-iky} dy$$

where k denotes the wave number associated with particle momentum.

Substituting the normalized Airy wavefunction,

$$\psi_n(y) = \frac{\sqrt{\alpha}}{|Ai'(-a_n)|} Ai(\alpha y - a_n)$$

gives

$$\phi_n(k) = \frac{\sqrt{\alpha}}{|Ai'(-a_n)|\sqrt{2\pi}} \int_0^\infty Ai(\alpha y - a_n) e^{-iky} dy.$$

The momentum probability density was then obtained from

$$P(k) = |\phi(k)|^2.$$

The Fourier transform therefore converts the quantum state from position space to momentum space and provides information regarding the momentum distribution of the particle. The position-space and momentum-space representations are complementary descriptions of the same quantum state and are related through the uncertainty principle.

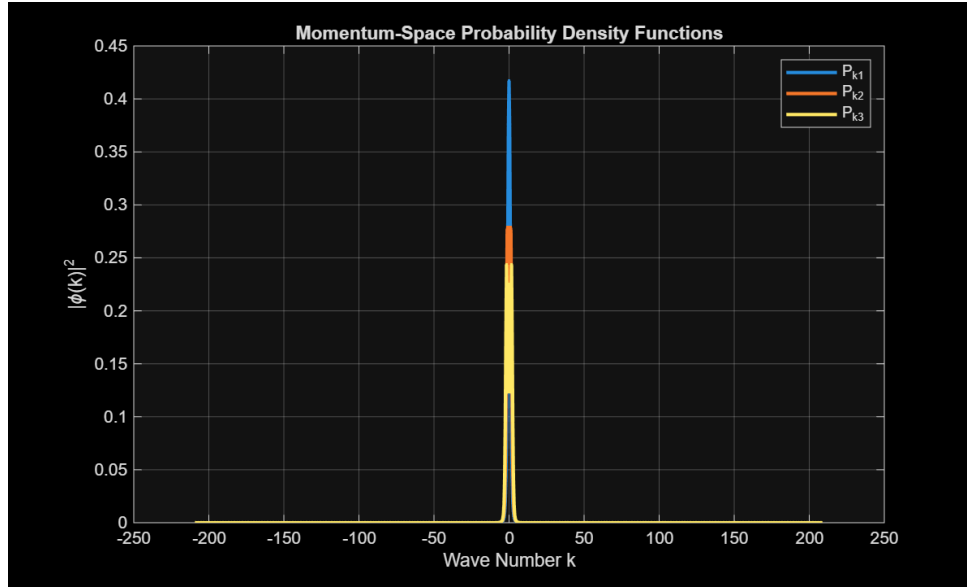


Figure 4.1: Fourier transform of the Airy-function wavefunction.

The Fourier transform provides a momentum-space representation of the quantum gravitational state. While the probability density function $|\psi(y)|^2$ describes the likelihood of finding the particle at a particular height, the momentum-space probability density $|\phi(k)|^2$ describes the likelihood of observing a particular momentum. The two representations contain the same physical information but in different domains. This dual representation illustrates the wave-particle nature of quantum systems and forms the basis of the position-momentum uncertainty principle.

Probability Theory and Random Variables

5.1 Random Variable Interpretation of the Quantum Gravitational System

In classical projectile motion, the position of the particle at any instant is deterministic and can be described exactly using Newtonian equations of motion. However, in quantum mechanics, the particle does not possess a single definite position. Instead, its position is described probabilistically through the wavefunction obtained from the Schrödinger equation.

For the gravitational quantum system, the normalized Airy-function wavefunction is given by:

$$\psi_n(y) = \frac{\sqrt{\alpha}}{|Ai'(-a_n)|} Ai(\alpha y - a_n)$$

where:

$$\alpha = \left(\frac{2m^2g}{\hbar^2} \right)^{1/3}$$

and a_n are the zeros of the Airy function.

5.1.1 Probability Density Function

After normalization, the wavefunction satisfies:

$$\int_0^\infty |\psi_n(y)|^2 dy = 1$$

which allows the squared magnitude of the wavefunction to be interpreted as a probability density function.

Thus, the probability density is defined as:

$$P(y) = |\psi_n(y)|^2$$

The quantity:

$$P(y) dy$$

represents the probability of finding the particle between heights y and $y + dy$.

Therefore, the vertical position of the particle was treated as a random variable:

$$Y \sim |\psi_n(y)|^2$$

This transformed the gravitational projectile system into a probabilistic quantum-mechanical model.

5.1.2 Physical Interpretation of the Probability Density

The probability density plots demonstrated the spatial distribution of the particle under gravity.

The regions where:

$$|\psi_n(y)|^2$$

attained large values corresponded to the most probable particle heights.

For higher energy states, the probability distributions extended over larger spatial regions and exhibited increased oscillatory behavior.

The wavefunction gradually decayed at larger heights due to the classically forbidden region where:

$$V(y) > E_n$$

As a result, the probability density also decreased exponentially for large values of y , indicating that the probability of finding the particle far above the surface became extremely small.

5.1.3 Expectation Value of Position

The expectation value of the random variable Y represents the average vertical position of the particle.

It is defined as:

$$\mathbb{E}[Y] = \int_0^\infty y |\psi_n(y)|^2 dy$$

The expectation value provides the mean height of the particle over many measurements.

The computed expectation values increased with increasing energy state, indicating that higher-energy particles occupied greater average heights against gravity.

Thus:

$$\mathbb{E}[Y]_1 < \mathbb{E}[Y]_2 < \mathbb{E}[Y]_3$$

This behavior was consistent with the probability density plots, where higher-energy states extended further upward.

5.1.4 Variance of Position

The variance of the random variable Y measures the spread of the probability distribution around the expectation value.

It is defined as:

$$\text{Var}(Y) = \mathbb{E}[Y^2] - \mathbb{E}[Y]^2$$

where:

$$\mathbb{E}[Y^2] = \int_0^\infty y^2 |\psi_n(y)|^2 dy$$

A small variance indicates that the particle is strongly localized around its mean position, while a large variance indicates that the particle is spatially spread over a larger region.

The computed variances increased with increasing energy state:

$$\text{Var}(Y)_1 < \text{Var}(Y)_2 < \text{Var}(Y)_3$$

This demonstrated that higher-energy Airy states were more spatially delocalized.

5.1.5 Relation to the Probability Density Graphs

The probability density plots visually confirmed the statistical behavior of the system.

- Lower energy states exhibited narrow and localized probability distributions.
- Higher energy states exhibited broader distributions with multiple peaks.
- The number of oscillatory peaks increased with the energy state.

This behavior indicated that higher-energy states contained larger spatial spreads and more complex standing-wave structures.

5.1.6 Connection to Quantum Mechanics and Computational Analysis

The random-variable formulation provided a statistical interpretation of the discretized quantum wavefunctions obtained from the computational eigenvalue analysis.

The probability densities were generated from the normalized eigenvectors of the Hamiltonian matrix:

$$H\Psi_n = E_n\Psi_n$$

Thus:

- Eigenvectors represented discretized wavefunctions
- Probability densities represented statistical particle distributions
- Expectation values represented average particle heights

- Variance and standard deviation quantified positional uncertainty

The addition of random variables therefore established a strong connection between:

- Quantum mechanics
- Probability theory
- Computational mechanics
- Eigenvalue analysis
- Intelligent systems mathematics

The gravitational projectile system was therefore interpreted not as a deterministic trajectory, but as a probabilistic quantum system characterized by quantized Airy-function eigenstates and statistical uncertainty.

Optimisation: Uncertainty Minimization

6.1 Optimization Through Uncertainty Minimization

In addition to the probabilistic analysis of the quantum gravitational system, an optimization framework was introduced to identify the quantum state with the highest degree of spatial localization.

The position of the particle was modeled as a random variable Y whose probability density function was obtained from the normalized wavefunction:

$$P(y) = |\psi(y)|^2$$

The spread of the probability distribution was quantified using the variance:

$$\text{Var}(Y) = \mathbb{E}[Y^2] - \mathbb{E}[Y]^2$$

where:

$$\mathbb{E}[Y] = \int_0^\infty y |\psi(y)|^2 dy$$

and

$$\mathbb{E}[Y^2] = \int_0^\infty y^2 |\psi(y)|^2 dy$$

The variance provides a measure of positional uncertainty. A small variance indicates that the probability density is concentrated around a narrow spatial region, whereas a large variance indicates that the particle is distributed over a wider range of heights.

6.1.1 Optimization Objective

The objective of the optimization process was to determine the quantum state that minimizes the positional uncertainty of the particle.

Mathematically, the optimization problem is expressed as:

$$n^* = \arg \min_n [\text{Var}(Y_n)]$$

where:

- n denotes the quantum state.
- $\text{Var}(Y_n)$ denotes the variance associated with the n^{th} state.
- n^* denotes the optimal state corresponding to minimum uncertainty.

The variances of the first three Airy-function states were computed and compared. The state possessing the smallest variance was identified as the most localized quantum state.

6.1.2 Determination of the Most Probable Height

After identifying the optimal state through variance minimization, the corresponding probability density function was analyzed to determine the height at which the probability of finding the particle was maximum.

The most probable height was obtained by solving:

$$y_{\text{peak}} = \arg \max_y P(y)$$

where:

$$P(y) = |\psi(y)|^2$$

In the numerical implementation, the probability density values were evaluated at discrete grid points and the maximum value was located using:

$$P_{\max} = \max(P)$$

The corresponding height was then obtained as:

$$y_{\text{peak}} = y(i_{\max})$$

where $i_{\max}$ denotes the index associated with the maximum probability density.

6.1.3 Physical Interpretation

The optimization process consisted of two stages:

1. Identification of the quantum state with minimum variance.
2. Determination of the most probable particle height within that optimal state.

The minimum-variance state represents the quantum configuration with the smallest positional uncertainty and therefore the strongest spatial localization.

The peak probability height represents the location where the particle is most likely to be observed during a measurement.

Consequently, the optimization framework identifies the quantum state that simultaneously exhibits the highest localization and the greatest confidence in particle position.

6.1.4 Optimization Results

The uncertainty minimization procedure was applied to the first three quantum states. The variance of each state was computed and compared to determine the most spatially localized state.

The optimization results are summarized in Table 6.1.

Table 6.1: Results of Uncertainty Minimization

Quantity	Value
Optimal State	1
Minimum Variance	0.306119
Most Probable Height	1.051051
Maximum Probability	0.735205

The results indicate that the first quantum state possesses the smallest variance and therefore represents the most localized quantum state. Since variance measures the spread of the probability distribution, the minimum variance corresponds to the lowest positional uncertainty.

After identifying the optimal state, the probability density function was analyzed to determine the most probable particle position. The maximum probability density was found to occur at a height of:

$$y_{\text{peak}} = 1.051051$$

with a corresponding probability density of:

$$P_{\text{max}} = 0.735205$$

This height represents the location where the particle is most likely to be observed during a measurement while simultaneously exhibiting the smallest positional uncertainty.

The optimization therefore identifies the quantum state that provides the strongest particle localization and the greatest confidence in particle position.

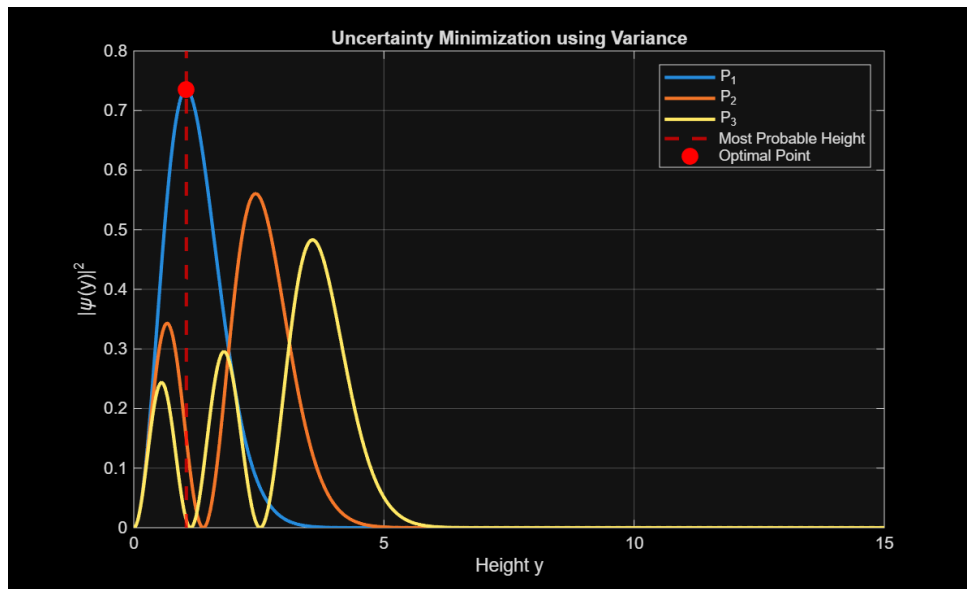


Figure 6.1: Uncertainty minimization using variance. The probability density functions of the first three quantum states are shown. The red dashed line indicates the most probable height corresponding to the minimum-variance state. The red marker identifies the optimal particle location. State 1 was found to possess the smallest variance and therefore represents the most spatially localized quantum state.

Results and Discussion

7.1 Introduction

The quantum gravitational projectile system was analyzed using both analytical and computational approaches. The Schrödinger equation under the linear gravitational potential was transformed into the Airy differential equation, and the resulting Airy-function solutions were studied analytically. Numerical computations were performed using MATLAB to visualize the wavefunctions, probability densities, and statistical properties of the quantum states.

7.2 Airy Wavefunctions

The normalized Airy-function wavefunctions for the first three quantum states were computed and plotted.

The wavefunctions exhibited oscillatory behavior near the lower region and decayed gradually at larger heights.

The following important observations were made:

- All wavefunctions satisfied the boundary condition:

$$\psi(0) = 0$$

due to the infinite potential wall at $y = 0$.

- The wavefunctions oscillated in the classically allowed region where:

$$E_n > V(y)$$

- The wavefunctions decayed exponentially in the classically forbidden region where:

$$V(y) > E_n$$

- Higher-energy states exhibited larger numbers of oscillations and nodes.
- The oscillatory structure indicated standing-wave-like quantum behavior.

The negative regions of the wavefunction did not represent negative probability. Instead, they represented phase changes and oscillatory behavior of the quantum state. The probability density remained positive because it depended on:

$$|\psi(y)|^2$$

The increasing oscillatory behavior for higher-energy states indicated increasing spatial frequency and more complex quantum eigenmodes.

7.3 Probability Density Distributions

The probability density functions were obtained from:

$$P(y) = |\psi(y)|^2$$

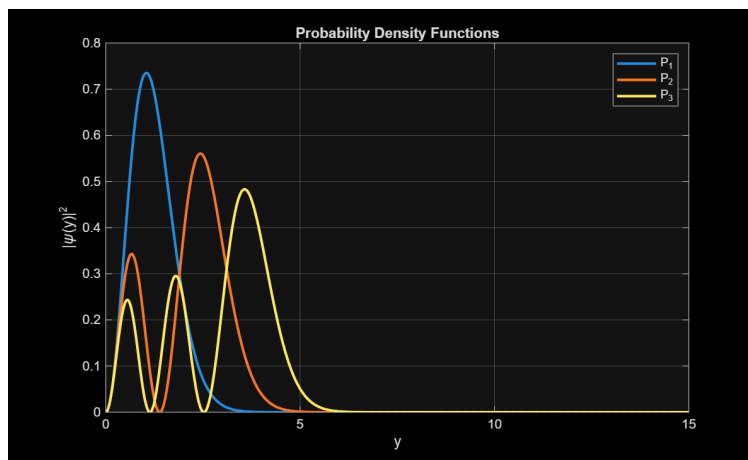


Figure 7.1: Probability density functions for the first three quantum states of the gravitational system.

These probability distributions represented the likelihood of finding the particle at different heights.

The probability density plots revealed several important physical properties:

- Lower-energy states were strongly localized near the surface.
- Higher-energy states extended further upward against gravity.
- The number of probability peaks increased with increasing quantum state.
- The probability density gradually approached zero at large heights due to exponential decay of the Airy function.

The peaks in the probability density represented the most probable particle positions.

The increasing number of peaks for higher-energy states demonstrated the standing-wave nature of the quantized gravitational system.

7.4 Comparison of Analytical and Computational Energy Eigenvalues

The numerical energy eigenvalues obtained from the discretized Hamiltonian matrix were compared with the analytical Airy-function energy eigenvalues.

The comparison is shown in Table 7.1.

Table 7.1: Comparison of Numerical and Analytical Energy Eigenvalues

Quantum State	Numerical Energy	Analytical Energy
1	8.610200	8.613634
2	15.049540	15.059952
3	20.318579	20.338064
4	24.973205	25.002417
5	29.226350	29.266315

The numerical results show excellent agreement with the analytical Airy-function energies. The small differences arise due to discretization error introduced by the finite difference approximation.

As the number of discretization points increases and the grid spacing decreases, the numerical eigenvalues converge toward the analytical solutions, validating the computational finite difference formulation.

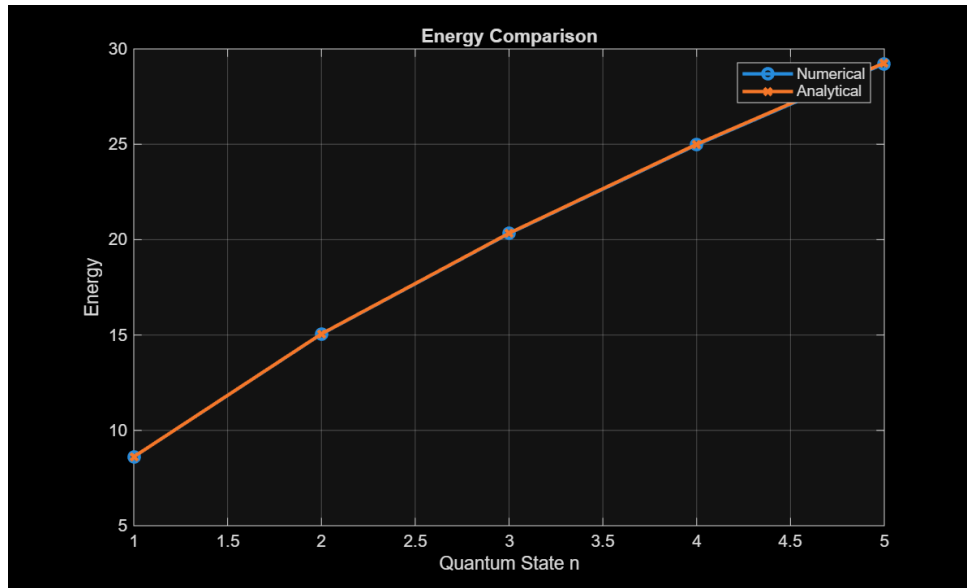


Figure 7.2: Comparison between analytical Airy-function energy eigenvalues and computational eigenvalues obtained from the discretized Hamiltonian matrix. The close agreement between the two sets of results validates the numerical finite difference implementation.

7.5 Expectation Values of Position

The expectation value of the vertical position was calculated using:

$$\mathbb{E}[Y] = \int_0^\infty y |\psi(y)|^2 dy$$

The computed expectation values were:

Quantum State	Expectation Value
State 1	1.237
State 2	2.163
State 3	2.921

The expectation value represented the average vertical height of the particle over many measurements.

The results showed that the expectation value increased with increasing energy state:

$$\mathbb{E}[Y]_1 < \mathbb{E}[Y]_2 < \mathbb{E}[Y]_3$$

This indicated that higher-energy quantum states occupied larger average heights against gravity.

The probability density plots visually confirmed this behavior, as the distributions shifted upward for higher-energy states.

7.6 Variance and Positional Uncertainty

The variance was computed using:

$$\text{Var}(Y) = \mathbb{E}[Y^2] - \mathbb{E}[Y]^2$$

The calculated variances were:

Quantum State	Variance
State 1	0.306
State 2	0.936
State 3	1.707

The standard deviations were:

Quantum State	Standard Deviation
State 1	0.553
State 2	0.967
State 3	1.307

The variance and standard deviation quantified the spread and uncertainty of the particle position.

The results demonstrated that positional uncertainty increased with increasing energy state.

Higher-energy states possessed broader probability distributions and larger spatial spreads.

This behavior indicated that higher-energy Airy states were more spatially delocalized.

7.7 Classically Allowed and Forbidden Regions

The computational results clearly demonstrated the distinction between the classically allowed and forbidden regions.

The classical turning point occurred at:

$$y_t = \frac{E_n}{mg}$$

In the classically allowed region:

$$E_n > V(y)$$

the wavefunctions oscillated.

In the classically forbidden region:

$$V(y) > E_n$$

the Airy wavefunctions decayed exponentially.

Although classical mechanics predicts that the particle cannot exist in the forbidden region, quantum mechanics allows a small but nonzero probability of existence in this region.

This demonstrated the probabilistic nature of the quantum gravitational system.

7.8 Computational Eigenvalue Analysis

The Schrödinger equation was discretized using the Finite Difference Method, converting the continuous differential equation into the matrix eigenvalue problem:

$$H\Psi_n = E_n\Psi_n$$

The eigenvalues represented the quantized energy levels, while the eigenvectors represented the discretized wavefunctions.

The numerical eigenvalues obtained from the Hamiltonian matrix were compared with the analytical Airy-function energy eigenvalues.

The computational results showed close agreement with the analytical solutions, validating the numerical finite difference formulation.

As the grid spacing decreased and the number of discretization points increased, the numerical energies converged toward the exact analytical Airy energies.

This confirmed the accuracy of the computational model.

7.9 Connection to Computational Mechanics and Intelligent Systems

The project established strong interdisciplinary connections between:

- Quantum mechanics
- Differential equations
- Computational mechanics
- Linear algebra
- Probability theory
- Eigenvalue analysis
- Intelligent systems mathematics

The discretized Schrödinger equation transformed the quantum system into a matrix eigenvalue problem, allowing the wavefunctions to be interpreted as computational eigenmodes.

The random-variable formulation introduced probabilistic analysis through expectation values, variance, and uncertainty quantification.

Thus, the projectile system was interpreted not as a deterministic classical trajectory, but as a probabilistic quantum-mechanical system governed by Airy-function eigenstates and statistical behavior.

7.10 Summary of Results

The major findings of the project are summarized below:

- The gravitational Schrödinger equation was successfully transformed into the Airy differential equation.
- Airy functions accurately described the quantized gravitational states.
- Boundary conditions produced discrete energy eigenvalues.
- Higher-energy states exhibited increased oscillations and probability peaks.
- Expectation values increased with increasing quantum state.
- Variance and standard deviation demonstrated increasing positional uncertainty for higher-energy states.

- The computational finite difference model accurately reproduced the analytical Airy-function results.
- The project successfully integrated quantum mechanics, computational mechanics, and probabilistic intelligent-system analysis into a unified framework.

Conclusions

8.1 Summary of Mathematical Themes

Table 8.1: Mathematical themes and their roles in the project.

Theme	Role in Project
Differential Equations	Governing Schrödinger ODE; reduction to Airy equation
Quantum Mechanics	Physical model; eigenstate interpretation; Born rule
Linear Algebra	Hamiltonian matrix construction; eigenvalue problem
SVD	Wavefunction matrix decomposition; mode extraction
Random Variables	Position PDF; Born rule probabilistic framework
Statistical Moments	Expectation, variance, standard deviation of height
Optimisation	Discrete maximisation of probability density at target

8.2 Key Conclusions

Principal Results

1. The Schrödinger equation under a linear gravitational potential reduces to the Airy equation, whose bounded solutions yield quantised energy eigenstates.
2. Discretisation of the wavefunction into a matrix and application of SVD extract the dominant spatial modes, demonstrating low-dimensional structure in the quantum state space.
3. The squared wavefunction constitutes a rigorous probability density function,

enabling a full probabilistic analysis of particle height.

4. Expectation values and variances increase monotonically with energy quantum number, confirming that higher states are more spatially delocalised.
5. The discrete optimisation over energy states successfully identifies the eigenstate that concentrates maximum probability at any prescribed target height.

8.3 Final Characterisation

A Probabilistic Eigenvalue- Based Model of Vertical Projectile Dynamics Under Gravity:

Airy-function eigenstates describe quantised gravitational motion, and probability distributions characterise the uncertainty and statistical behaviour of the particle's vertical position.

MATLAB Code Listings

A.1 Wavefunction Computation, Fourier Transform Random Variables and Optimization

```
1 clc; clear; close all;
2
3 %% PARAMETERS
4
5 hbar = 1;
6 m = 1;
7 g = 1;
8
9 %% DOMAIN
10
11 y = linspace(0,15,1000);
12
13 %% AIRY FUNCTION PARAMETER
14
15 alpha = (2*m^2*g/hbar^2)^(1/3);
16
17 %% FIRST THREE AIRY ZEROS
18
19 a1 = 2.3381;
20 a2 = 4.0879;
21 a3 = 5.5206;
22
23 %% WAVEFUNCTIONS
```

```

24
25 psi1 = airy(alpha*y - a1);
26 psi2 = airy(alpha*y - a2);
27 psi3 = airy(alpha*y - a3);
28
29 %% NORMALIZATION
30
31 N1 = sqrt(trapz(y,abs(psi1).^2));
32 N2 = sqrt(trapz(y,abs(psi2).^2));
33 N3 = sqrt(trapz(y,abs(psi3).^2));
34
35 psi1 = psi1/N1;
36 psi2 = psi2/N2;
37 psi3 = psi3/N3;
38
39 %% PROBABILITY DENSITIES
40
41 P1 = abs(psi1).^2;
42 P2 = abs(psi2).^2;
43 P3 = abs(psi3).^2;
44
45 %% EXPECTATION VALUES
46
47 Ey1 = trapz(y, y .* P1);
48 Ey2 = trapz(y, y .* P2);
49 Ey3 = trapz(y, y .* P3);
50
51 %% EXPECTATION OF y^2
52
53 Ey1_sq = trapz(y, y.^2 .* P1);
54 Ey2_sq = trapz(y, y.^2 .* P2);
55 Ey3_sq = trapz(y, y.^2 .* P3);
56
57 %% VARIANCE
58
59 Var1 = Ey1_sq - Ey1^2;
60 Var2 = Ey2_sq - Ey2^2;
61 Var3 = Ey3_sq - Ey3^2;
62
63 %% DISPLAY RESULTS

```

```

64
65 disp('-----')
66 disp(' STATE 1 ')
67 disp('-----')
68 fprintf('Expectation Value <y> = %f\n', Ey1)
69 fprintf('Variance = %f\n\n', Var1)
70
71 disp('-----')
72 disp(' STATE 2 ')
73 disp('-----')
74 fprintf('Expectation Value <y> = %f\n', Ey2)
75 fprintf('Variance = %f\n\n', Var2)
76
77 disp('-----')
78 disp(' STATE 3 ')
79 disp('-----')
80 fprintf('Expectation Value <y> = %f\n', Ey3)
81 fprintf('Variance = %f\n\n', Var3)
82
83 %% PLOT WAVEFUNCTIONS
84
85 figure;
86
87 plot(y,psi1,'LineWidth',2)
88 hold on
89 plot(y,psi2,'LineWidth',2)
90 plot(y,psi3,'LineWidth',2)
91
92 xlabel('Height y')
93 ylabel('\psi(y)')
94
95 title('Normalized Airy Wavefunctions')
96
97 legend('\psi_1', '\psi_2', '\psi_3')
98
99 grid on
100
101 %% PLOT PROBABILITY DENSITIES
102
103 figure;

```

```

104
105 plot(y,P1,'LineWidth',2)
106 hold on
107 plot(y,P2,'LineWidth',2)
108 plot(y,P3,'LineWidth',2)
109
110 xlabel('Height y')
111 ylabel('|\psi(y)|^2')
112
113 title('Probability Density Functions')
114
115 legend('P_1','P_2','P_3')
116
117 grid on
118
119 %% UNCERTAINTY MINIMIZATION
120
121 Variances = [Var1 Var2 Var3];
122
123 [minVar, optimalState] = min(Variances);
124
125 disp('-----')
126 disp(' UNCERTAINTY MINIMIZATION ')
127 disp('-----')
128 fprintf('Optimal State = %d\n', optimalState)
129 fprintf('Minimum Variance = %f\n\n', minVar)
130
131 %% FIND MOST PROBABLE HEIGHT OF OPTIMAL STATE
132
133 if optimalState == 1
134
135     [Pmax,idx] = max(P1);
136     y_peak = y(idx);
137
138 elseif optimalState == 2
139
140     [Pmax,idx] = max(P2);
141     y_peak = y(idx);
142
143 else

```

```

144     [Pmax,idx] = max(P3);
145     y_peak = y(idx);
146
147
148 end
149
150 fprintf('Most Probable Height = %f\n', y_peak)
151 fprintf('Maximum Probability = %f\n', Pmax)
152
153 %% OPTIMIZATION VISUALIZATION
154
155 figure;
156
157 plot(y,P1,'LineWidth',2)
158 hold on
159 plot(y,P2,'LineWidth',2)
160 plot(y,P3,'LineWidth',2)
161
162 % Most probable height of minimum variance state
163 xline(y_peak,'r--','LineWidth',2)
164
165 % Mark optimal point
166 plot(y_peak,Pmax,...
167      'ro',...
168      'MarkerSize',8,...
169      'LineWidth',2,...
170      'MarkerFaceColor','r')
171
172 xlabel('Height y')
173 ylabel('|\psi(y)|^2')
174
175 title('Uncertainty Minimization using Variance')
176
177 legend('P_1','P_2','P_3',...
178      'Most Probable Height',...
179      'Optimal Point')
180
181 grid on
182
183 %% FOURIER TRANSFORM (POSITION -> MOMENTUM SPACE)

```

```

184
185 dy = y(2) - y(1);
186
187 % Fourier transforms
188 phi1 = fftshift(fft(psi1));
189 phi2 = fftshift(fft(psi2));
190 phi3 = fftshift(fft(psi3));
191
192 % Corresponding wave-number axis
193 N = length(y);
194
195 k = (-N/2:N/2-1)*(2*pi/(N*dy));
196
197 % Momentum-space probability densities
198 Pk1 = abs(phi1).^2;
199 Pk2 = abs(phi2).^2;
200 Pk3 = abs(phi3).^2;
201
202 % Normalize momentum-space PDFs
203 Pk1 = Pk1/trapz(k,Pk1);
204 Pk2 = Pk2/trapz(k,Pk2);
205 Pk3 = Pk3/trapz(k,Pk3);
206
207 %% PLOT MOMENTUM DISTRIBUTIONS
208
209 figure;
210
211 plot(k,Pk1,'LineWidth',2)
212 hold on
213
214 plot(k,Pk2,'LineWidth',2)
215
216 plot(k,Pk3,'LineWidth',2)
217
218 xlabel('Wave Number k')
219 ylabel('|phi(k)|^2')
220
221 title('Momentum-Space Probability Density Functions')
222
223 legend('P_{k1}','P_{k2}','P_{k3}')

```

```

224
225 grid on

```

A.2 Wavefunction Matrix

```

1  clc; clear; close all;
2
3  %% PARAMETERS
4
5  hbar = 1;          % Reduced Planck constant
6  m    = 1;          % Mass
7  g    = 10;         % Gravity
8
9  L = 15;            % Maximum height/domain
10 N = 400;           % Number of grid points
11
12 dy = L/(N+1);
13
14 y = linspace(dy, L-dy, N)';
15
16 %% FINITE DIFFERENCE HAMILTONIAN
17
18 % Kinetic coefficient
19 t = hbar^2/(2*m*dy^2);
20
21 % Potential V(y) = mgy
22 V = m*g*y;
23
24 % Main diagonal
25 main_diag = 2*t + V;
26
27 % Off diagonals
28 off_diag = -t*ones(N-1,1);
29
30 % Hamiltonian matrix
31 H = diag(main_diag) + ...
32     diag(off_diag,1) + ...
33     diag(off_diag,-1);
34

```

```

35 %% SOLVE EIGENVALUE PROBLEM
36
37 % Eigenvectors and eigenvalues
38 [psi,E] = eig(H);
39
40 % Extract eigenvalues
41 E_values = diag(E);
42
43 % Sort energies
44 [E_values,idx] = sort(E_values);
45
46 % Reorder eigenvectors
47 psi = psi(:,idx);
48
49 %% ANALYTICAL AIRY ENERGIES
50
51 airy_zeros = [2.3381 4.0879 5.5206 6.7867 7.9441];
52
53 alpha = (hbar^2 * m * g^2 / 2)^(1/3);
54
55 E_exact = airy_zeros * alpha;
56
57 %% DISPLAY RESULTS
58
59 disp('-----');
60 disp(' Numerical vs Analytical Energies ');
61 disp('-----');
62
63 for n = 1:5
64
65     fprintf('n = %d\n',n);
66     fprintf('Numerical Energy = %f\n',E_values(n));
67     fprintf('Analytical Energy = %f\n\n',E_exact(n));
68
69 end
70
71 %% PLOT ENERGY COMPARISON
72
73 figure;
74

```



```
75 plot(1:5,E_values(1:5),'o-','LineWidth',2)
76 hold on
77
78 plot(1:5,E_exact,'x-','LineWidth',2)
79
80 xlabel('Quantum State n')
81 ylabel('Energy')
82
83 legend('Numerical','Analytical')
84
85 title('Energy Comparison')
86
87 grid on
```

List of Symbols

Symbol	Meaning	SI Unit
$\hbar$	Reduced Planck constant	J·s
m	Particle mass	kg
g	Gravitational acceleration	m s ⁻²
y	Vertical position coordinate	m
$\psi_n(y)$	n -th energy eigenfunction	m ^{-1/2}
E_n	n -th energy eigenvalue	J
z	Dimensionless position variable	—
a_n	n -th zero of Ai	—
Ai(z)	Airy function of the first kind	—
Bi(z)	Airy function of the second kind	—
$P_n(y)$	Probability density function of Y_n	m ⁻¹
Y_n	Vertical position random variable, state n	m
$\mathbb{E}[Y_n]$	Expectation value of Y_n	m
$\text{Var}(Y_n)$	Variance of Y_n	m ²
H	Discrete Hamiltonian matrix	J
Ψ	Wavefunction matrix	m ^{-1/2}
y^*	Target height for optimisation	m
n^*	Optimal energy state index	—
